

Data Analysis in R

Hypothesis Testing

Ken Stiller

10th June 2026

Syllabus: Data Analysis in R

1. Introduction
2. Causality & Basics of Statistics
3. Sampling & Measurement
4. Prediction
5. Multivariate Regression
6. Probability & Uncertainty
7. **Hypothesis Testing**
8. Assumptions & Limits of OLS
9. Interactions & Non-Linear Effects

Plan for Today

- ▶ **Accuracy**
 - ▶ Standard Errors
 - ▶ Confidence Intervals
- ▶ **Hypothesis Testing**
 - ▶ What is a hypothesis?
 - ▶ p-values
 - ▶ Type I and Type II errors
 - ▶ One and two-sided tests

Table of Contents

Overview

Statistical Inference

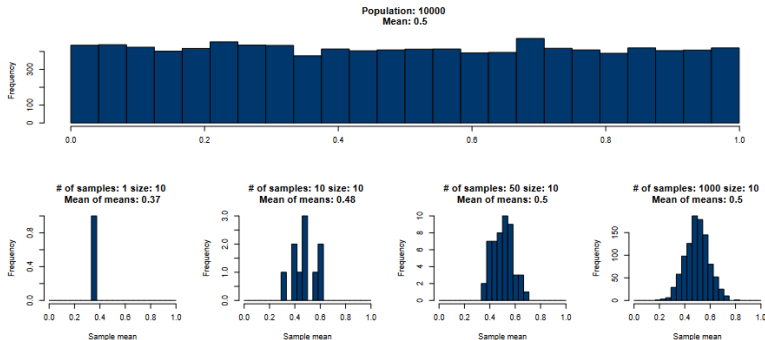
Hypothesis Testing

Wrap Up

Recap: Statistical Inference

- ▶ What we want to know: **parameter** $\theta \rightsquigarrow$ unobservable
- ▶ What you do observe: **data**
- ▶ We use data to compute an **estimate** of the parameter ($\hat{\theta}$)
- ▶ **But how good is $\hat{\theta}$ an estimate of θ ?**
- ▶ Ideally, we want to know the **estimation error** $= \hat{\theta} - \theta$
- ▶ The problem remains unchanged: θ is unknown

Recap: Central Limit Theorem



Confidence Intervals

- ▶ We know the standard error and are aware of the Central Limit Theorem
- ▶ Thus, we can calculate specific ranges around the sample mean of which, if repeated over and over again, a certain share of will contain the population mean. In other words, we can quantify how *confident* we are in our estimate.
- ▶ This is called a **confidence interval**

Confidence Intervals II

- ▶ An m -percent confidence interval establishes a boundary around the sample mean in which the true mean will lie m out of 100 times under repeated sampling
- ▶ Common values for m are 95 and 99 (sometimes 90)
- ▶ m is specified by choosing a significance level α : $m = (1 - \alpha) * 100$
- ▶ Common significance levels are therefore 0.05 and 0.01 (sometimes 0.1)

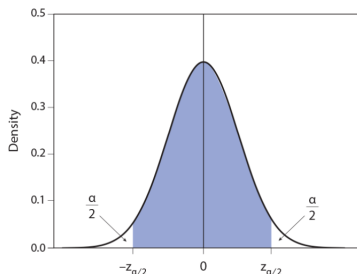
Confidence Intervals: Defining Boundaries

- ▶ In order to provide an interval estimate of the population mean μ we need to identify a lower bound (LB) and an upper bound (UB) such that $P(LB \leq \mu \leq UB)$
- ▶ Recall last session on probability: We can use our knowledge of the normal distribution to find this boundary
- ▶ After z-transformation of any normal distribution

$$z = \frac{x_i - \bar{x}}{s_x}$$

- ▶ Probability between -1 and 1 is 0.68
 - ▶ Probability between -1.96 and 1.96 is 0.95
 - ▶ Probability between -3 and 3 is 0.997
- ▶ $z_{\alpha/2}$ is the value associated with $(1 - \alpha) * 100\%$ coverage in the standard normal distribution

Example: Critical Values of Normal Distribution



- The lower and upper critical values, $-z_{\alpha/2}$ and $z_{\alpha/2}$, shown on horizontal axis
- Area under the density curve between these critical values (in blue) equals $1 - \alpha$

Confidence Intervals: Overview

- ▶ CI: boundaries in which μ will lie m -times out of a 100
- ▶ $(1 - \alpha) * 100\%$ confidence intervals:

$$CI_{\alpha} = [\bar{X} - z_{\alpha/2} * SE, \bar{X} + z_{\alpha/2} * SE]$$

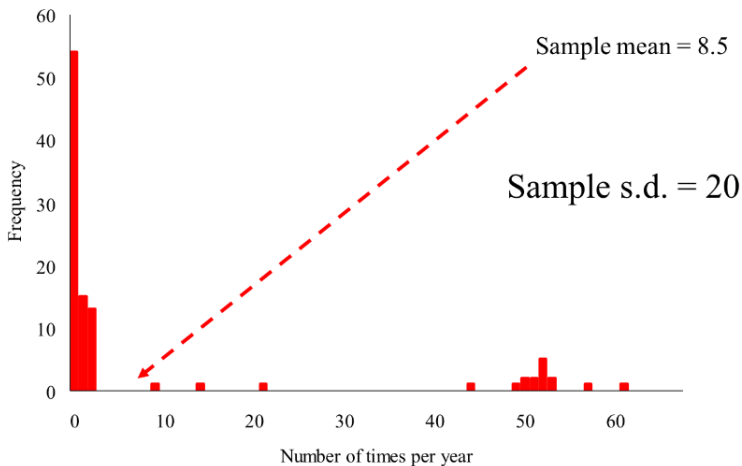
where $z_{\alpha/2}$ is the **critical value** and α reflects our chosen significance level

- ▶ $P(Z > z_{\alpha/2}) = \alpha/2$ and $Z \sim \mathcal{N}(0, 1)$
 1. $\alpha = 0.01$ gives $z_{\alpha/2} = 2.58$
 2. $\alpha = 0.05$ gives $z_{\alpha/2} = 1.96$
 3. $\alpha = 0.10$ gives $z_{\alpha/2} = 1.64$

Example: Confidence Intervals & Standard Error for Sample Mean

- ▶ How often do German people attend some form of religious worship?
- ▶ Take a random sample of 100 people from the German population and record how many times they attended a form of religious worship last year
 - ▶ Distribution is extremely skewed
 - ▶ Some went a lot, most went infrequently or not at all
- ▶ From that sample we get a sample mean and a sample standard deviation

Example: Confidence Intervals & Standard Error for Sample Mean II



Example: Confidence Intervals & Standard Error for Sample Mean III

- ▶ From the *sd* and *mean* of the sample we can calculate the **standard error** for the sample mean:

$$\frac{s}{\sqrt{n}} = \frac{20}{\sqrt{100}}$$

where s = sample standard deviation

- ▶ From this we can calculate any **confidence interval**

$$CI_{\alpha} = [\bar{X} - z_{\alpha/2} \times \text{standard error}, \bar{X} + z_{\alpha/2} \times \text{standard error}]$$

- ▶ Usually, we are interested in a 95% CI:

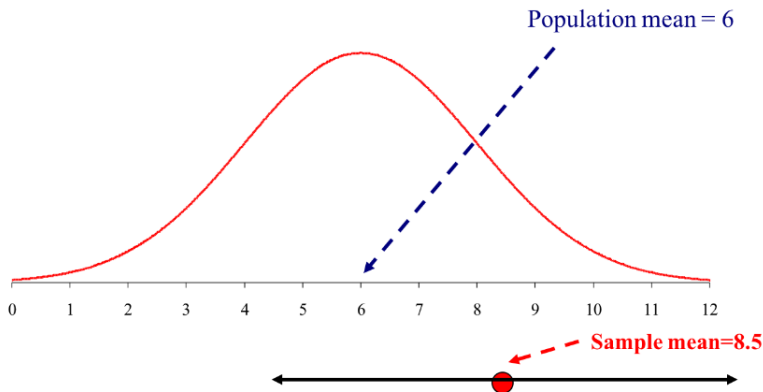
$$[8.5 - 1.96 \times 2, 8.5 + 1.96 \times 2] = [4.58, 12.42]$$

- ▶ In 95 out of a 100 times will the true mean lie within confidence intervals computed in this way

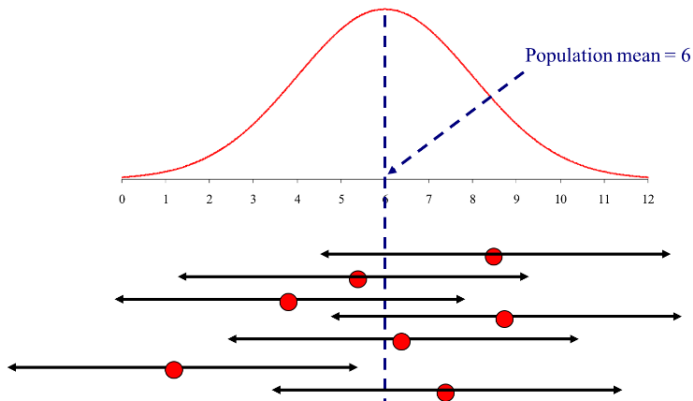
Example: Confidence Intervals & Standard Error for Sample Mean IV

- ▶ Suppose we know that the population mean for religious worship attendance in Germany is actually 6 times per year
 - ▶ Our particular sample is off by 2.5
 - ▶ The mean of all the possible sample means is equal to the population mean so the centre of the sampling distribution is 6

Example: Confidence Intervals & Standard Error for Sample Mean



Example: Confidence Intervals & Standard Error for Sample Mean II



Example: Confidence Intervals & Standard Error for Sample Mean VII

- ▶ Of the 7 samples, all the confidence intervals around the sample mean enclosed the actual true population mean apart from one
- ▶ If we repeated this lots of times, we would expect 95% of the confidence intervals to enclose the actual population mean
 - ▶ 95% because that's the level we set
 - ▶ If we had set 99%, the confidence intervals would be larger

Statistical Hypothesis Testing: Overview

1. Construct a **null hypothesis** (H_0) and its **alternative** (H_1)
2. Pick a **test statistic** T
3. Figure out the sampling distribution of T under H_0 (**reference distribution**)
 - ▶ For hypothesis tests regarding the mean, if sample size large, use the normal distribution
 - ▶ For other test statistics, you need to use other distributions
4. Is the observed value of T likely to occur under H_0 ?
 - ▶ **Yes** - Retain H_0
 - ▶ **No** - Reject H_0

What is a Hypothesis?

- ▶ Hypotheses = **testable statements** about the world
- ▶ Hypotheses = **falsifiable**
 - ▶ We test hypotheses by attempting to see if they could be false, rather than ‘proving’ them to be true
- ▶ Hypotheses come from:
 - ▶ Theory
 - ▶ Past empirical work
 - ▶ Common sense (?)
 - ▶ Anecdotal observations

Null and Alternative Hypotheses

- ▶ We choose between two conflicting statements, doing the following:
 1. The **null hypothesis** (H_0) is directly tested
 - ▶ This is a statement that the parameter we are interested in has a value similar to **no effect** (i.e., usually 0 for coefficients)
 - ▶ e.g. regarding ideology, old people are the same as young people
 2. **Alternative** (H_1) contradicts the null hypothesis
 - ▶ This is a statement that the parameter falls into a different set of values than those predicted by (H_0)
 - ▶ e.g. regarding ideology, old people are more right-wing than young people
- ▶ Note that we actually 'test' the null hypothesis!

Hypothesis Testing: Test Statistic

- ▶ In any statistical hypothesis test, a **test statistic** is computed from the data in order to test the null hypothesis.

$$T = \frac{\text{sample estimate} - \text{parameter value under } H_0}{\text{standard error}}$$

- ▶ The larger T , the more the data contradict the null hypothesis
- ▶ For a given estimate, T becomes larger as the standard error decreases

Statistical Hypothesis Testing: Overview II

- ▶ Hypotheses - $H_0: \mu = \mu_0$ and $H_1: \mu \neq \mu_0$
- ▶ Test statistic:

$$\text{z-score} = \frac{\bar{X} - \mu_0}{\text{standard error}} = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

- ▶ Under the null, by the central limit theorem

$$\text{z-score} \overset{\text{approx.}}{\sim} \mathcal{N}(0, 1)$$

- ▶ Is Z_{obs} unusual under the null?
 - ▶ Reject the null when $|Z_{obs}| > z_{\alpha/2}$
 - ▶ Retain the null when $|Z_{obs}| \leq z_{\alpha/2}$

Example: Exam Scores

- ▶ Suppose there's a standardised exam with marks ranging from 0-100
- ▶ Suppose further we know test scores are normally distributed with mean $\mu = 88$ and standard deviation $\sigma = 5$
- ▶ Now, in five tests cohorts receive test scores of $\bar{X} = 95$
 $H_0 : \mu = 88$ $H_1 : \mu \neq 88$

Example: Exam Scores II

- ▶ We know that the standard deviation of test in the population is 5. The sample size is 5 so we calculate the standard error as:

$$SE = \frac{5}{\sqrt{5}}$$

- ▶ Assuming H_0 was true, we know that the sampling distribution is

$$\mathcal{N}(88, (\frac{5}{\sqrt{5}})^2)$$

- ▶ Based on sampling distribution, how many standard deviations away is the observed mean from the hypothesized mean?

$$z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{95 - 88}{\frac{5}{\sqrt{5}}} = 5.82$$

- ▶ What is the probability of observing a z-value of more than 5.82 or below -5.82?

Example: Exam Scores III

The probability is smaller than 0.00001. What do we make of this?

p-Value

- ▶ Ok, so what's a p-value then?
- ▶ A **p-value** indicates the probability, under H_0 , of observing a value of the test statistic at least as extreme as its observed value
- ▶ A smaller p-value presents stronger evidence against H_0
- ▶ The level of the test: $\Pr(\text{rejection}|H_0) = \alpha$
- ▶ A p-value less than α conventionally indicates **statistical significance**
 - ▶ Conventional values of α : 0.05 & 0.01

p-Value II

- ▶ A **p-value** is an *arbitrary* threshold that our test must meet
 - ▶ we might want to be 99% confident that we are correctly rejecting the null hypothesis
 - ▶ or we might make the judgement that p-values of e.g. 0.05 and below are **probably** good evidence the null hypothesis can be rejected
- ▶ Keep in mind: the p-value is **not** the probability that H_0 (H_1) is true (false)

Type I and Type II Errors

- ▶ Concern false rejection if the null is true (*type I error*)
- ▶ Two types of errors:

	Reject H_0	Retain H_0
H_0 is true	Type I error	Correct
H_0 is false	Correct	Type II error

- ▶ Type I error occurs when we reject H_0 even though it is true
 - ▶ Happens 5% of the time if we choose $\alpha = 0.05$
- ▶ Type II error occurs when we do not reject H_0 even though it is false
 - ▶ If $\alpha = 0.05$, sometimes a real difference won't be detected

Type I and Type II Errors

- ▶ There's a trade-off between the two types of error
 - ▶ What probability do you want to minimize? False positive or false negative?

One- or Two-Sided (Tailed) Tests

- ▶ In the example above, we were interested in the difference to the true value
 - ▶ one-sided alternative hypothesis: $H_1 : \mu > \mu_0$ or $\mu < \mu_0$
 - ▶ one-sided p-value = $\Pr(Z > Z_{obs})$ or $\Pr(Z < Z_{obs})$
- ▶ Convention is to use two-tailed tests
 - ▶ making it even more difficult to find results just due to chance
 - ▶ normally don't have very strong prior information about the difference

Significance Tests and CIs

- ▶ Note that our significance test looks similar to the CIs
- ▶ We could use a CI around the difference between the two sample means to 'test' the hypothesis that they are the same
- ▶ A 95% CI would just be $1.96 * SE$
- ▶ You can see this on first view if the 95% CI encloses zero
- ▶ CIs and significance tests are doing the same job, just presenting the information in a slightly different way

Binary Variables and Proportions

- ▶ We have been working with continuous variables and means
- ▶ This works for binary variables too, where the mean is just the proportion:
 - ▶ Population mean = μ = population proportion = π
 - ▶ Population standard deviation = $\sigma = \sqrt{\pi(1 - \pi)}$
 - ▶ Sample proportion = P
 - ▶ Standard deviation (P) = $\frac{\sqrt{\pi(1-\pi)}}{\sqrt{n}}$
 - ▶ Standard error (P) = $\frac{\sqrt{P(1-P)}}{\sqrt{n}} = \sqrt{\frac{P(1-P)}{n}}$

Key Take Aways

- ▶ Knowing the shape of the sampling distribution, we can work out:
 - ▶ ranges around a sample mean that will enclose the population mean $X\%$ of the time
 - ▶ the probability that a null hypothesis about the population mean is 'true', given a particular sample mean
 - ▶ the probability that population means for different groups are different, given two sample means
 - ▶ all of the above for proportions
- ▶ Note that this allows us to make a **probabilistic** statement. Not more, not less.
- ▶ In expectation, a (non-negligible) share will be false positives!

The way ahead

- ▶ What if OLS assumptions are violated?
- ▶ When do we care?
- ▶ What can we do?
- ▶ What do we actually do?